

Program Content

Semester	VI		
Course Code:	IT6306		
Course Name:	Mobile Application Development		
Credit Value:	4 (2L + 2P)		
Core/Optional	Core		
Hourly Breakdown	Theory	Practical	Independent Learning
	30 Hrs	60 Hrs	110 Hrs
Course Aims: • Develop an understanding of contemporary mobile development platforms and skills required to develop applications for mobile devices • Explore a range of technical problems and solutions inherent in developing software applications for mobile devices including connectivity, security, and data storage. • Explain the key challenges in creating usable and effective interactive mobile applications and design techniques to address them • Develop an understanding of the unique features of contemporary mobile devices and how they can be used in interactive mobile application			
Intended Learning Outcomes: After following this course, students should be able to LO1: Explain the major milestones in the evolution of mobile devices LO2: Compare multiple mobile application development approaches LO3: Develop interactive mobile applications using a modern mobile development environment. LO4: Identify solutions to problems relating to interactive mobile applications. LO5: Design a moderately complex native mobile application in Android. LO6: Develop mobile applications for distribution on the Google Play Store.			
Course Content: (Main Topics, Subtopics)			
Topic	Theory (Hrs.)	Practical (Hrs.)	
1. Introduction to Mobile Applications	02	-	
2. Comparison of Mobile Application Development Platforms	03	-	
3. Designing for Mobile Applications	04	08	
4. Native Application Development with Android	12	30	
5. Android Architecture Components and Room Database	06	14	
6. Deployment and Monetization	03	08	
Total	30	60	

1. Introduction to mobile applications (02 hours)

- 1.1. The term “Mobility” in general [Ref 1: Pg. (19-28)]
- 1.2. History of Mobile devices [Ref 3: pg 1-3]
 - 1.2.1. The brick era [Ref 3: pg 3-4]
 - 1.2.2. The candy bar era [Ref 3: pg 5]
 - 1.2.3. The feature phone era [Ref 3: pg 6-7]
 - 1.2.4. The smart phone era [Ref 3: pg 8-9]
 - 1.2.5. The touch era [Ref 3: pg 10-12]
- 1.3. Layers of mobile eco system [Ref 3: pg 13]
 - 1.3.1. Operators [Ref 3: pg 14-16]
 - 1.3.2. Networks [Ref 3: pg 17]
 - 1.3.3. Devices [Ref 3: pg 18-19]
 - 1.3.4. Platforms [Ref 3: pg 20-21]
 - 1.3.5. Operating systems [Ref 3: pg 13]
 - 1.3.6. Application Frameworks [Ref 3: pg 22-25]
 - 1.3.7. Applications [Ref 3: pg 25]
 - 1.3.8. Services [Ref 3: pg 26]
- 1.4. Developing a mobile strategy [Ref 3: pg 57-67]

2. Mobile application development platforms (03 hours)

- 2.1. Introduction to mobile application development platforms [Ref 2: pg.109-110]
- 2.2. Android development platform [Ref 2: pg.151-153]
- 2.3. iOS development platform [Ref 2: pg.183-187]
- 2.4. Selecting the proper development platform [Ref 2: pg. 2-6]

3. Design for mobile applications (04 hours)

- 3.1. Introduction to mobile design [Ref 3: pg 109-115]
- 3.2. Elements of mobile design [Ref 3: Chap 8, 4]
 - 3.2.1. Context [Ref 3: pg 116]
 - 3.2.2. Message [Ref 3: pg 117]
 - 3.2.3. Look and feel [Ref 3: pg 118-120]
 - 3.2.4. Layout [Ref 3: pg 121-124]
 - 3.2.5. Color [Ref 3: pg 125-128]
 - 3.2.6. Typography [Ref 3: pg 129-133]
 - 3.2.7. Graphics [Ref 3: pg 134-136]
- 3.3. Popular prototyping platforms (e.g., Proto.io, Figma, etc.) [Ref 4]

4. Native Application Development with Android (12 hours)

- 4.1. Getting started with Android development
 - 4.1.1. Introduction to Android development [Ref 1]
 - 4.1.2. Setting up the tools and environment [Ref 1]
- 4.2. Working with User Interfaces
 - 4.2.1. App manifest and resources [Ref 1]
 - 4.2.2. Activities and Fragments (Activity Life Cycle, Fragment Life Cycle) [Ref 1]
 - 4.2.3. Layouts, Adapters, Action bar, Dialogs and Notifications [Ref 1]
- 4.3. Data and App interaction
 - 4.3.1. Intents and Broadcast Receivers [Ref 1]

- 4.3.2.Preferences and Saving State [Ref 1]
- 4.3.3.Content Providers and Services [Ref 1]
- 4.3.4. AsyncTask and AsyncTaskLoader [Ref 1]
- 4.4. Sensors and Communication
 - 4.4.1.Sensors (Sensor Identification and Registration) [Ref 1]
 - 4.4.2.Orientation and Movement (Pitch, roll and yaw, Natural Device Orientation, Reference frame remapping) [Ref 1]
 - 4.4.3.Sending and Receiving SMS [Ref 1]

5. Android Architecture Components and Room Database (06 hours)

- 5.1. Introduction to Android Architecture Components
 - 5.1.1.Activity / Fragment [Ref 1]
 - 5.1.2.ViewModel [Ref 1]
 - 5.1.3.Repository [Ref 1]
- 5.2. Room Database
 - 5.2.1.Room Overview [Ref 1]
 - 5.2.2.Components of Room
 - 5.2.2.1. Entity [Ref 1]
 - 5.2.2.2. DAO (Data Access Object) [Ref 1]
 - 5.2.2.3. Database [Ref 1]
- 5.3. Lifecycle-aware Components
 - 5.3.1.Usecases and Lifecycle library [Ref 1]
 - 5.3.2.Lifecycle Events and Observers [Ref 1]
 - 5.3.3.LiveData [Ref 1]

6. Deployment and Monetization (03 hours)

- 6.1. Deploying the Android App
 - 6.1.1. App releasing strategies [Ref 1]
 - 6.1.2.Prepare for release [Ref 1]
 - 6.1.3. Versioning the app [Ref 1]
 - 6.1.4. Sign the app [Ref 1]
 - 6.1.5. Uploading the app [Ref 1]
 - 6.1.6. Choosing the right monetization strategy [Ref 1]
 - 6.1.7. Google Play's subscription platform [Ref 1]
 - 6.1.8. Using Google AdMob [Ref 1]

Teaching /Learning Methods:

You can access all learning materials and this syllabus in the VLE: <http://vle.bit.lk/> if you are a registered student of the BIT degree program.

Assessment Strategy:

In the course, case studies/Lab sheets will be introduced, and students have to participate in the learning activities.

The final exam of the course will be held at the end of the semester. This course is evaluated using a two-hour question paper consisting of 4 Structured Questions.

References/ Reading Materials:

- **Ref 1.** Android Official Documentation. 2021. [online]
Available at: <<https://developer.android.com/>>
- **Ref 2.** McWherter, J., & Gowell, S. (2012). Professional mobile application development. John Wiley & Sons, Incorporated.
- **Ref 3.** Brian Fling, (2009). Mobile Design and Development (Practical techniques for creating mobile sites and web apps. O'Reilly Media, Inc.

Supplementary resources:

Ref 4. <https://help.figma.com/hc/en-us/articles/360040314193-Guide-to-prototyping-in-Figma>